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Natural Language Processing Applications



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Session 6: Sentiment Analysis

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Outline



- What is Sentiment Analysis
 - Sentiment Analysis approaches
 - Aspect based sentiment analysis
 - Tools and Technology stack
 - Production environment issues and challenges
 - Evaluation measures
 - Code demo
-



Motivation For Sentiment Analysis

What others think has always been an important piece of information

“Which car should I buy?”

*“Which schools should I
apply to?”*

“Which Professor to work for?”

“Whom should I vote for?”

What is Sentiment analysis

- **Computational study of opinions**, sentiments, evaluations, attitudes, appraisal, affects, views, emotions, subjectivity, etc., **expressed in text**.



The problem is..



- “Whoala! I have the reviews I need”
- *Now that I have “too much” information on one topic...I could easily form my opinion and make decisions...*
 - Is this true?
 - ...Not Quite
- Searching for reviews may be difficult
 - Can you search for opinions as conveniently as general Web search?
 - eg: is it easy to search for “*iPhone vs Google Phone*”?

Facts and Opinions

Two main types of information on the Web.

- Facts(Objective) and Opinions(Subjective)

Fact : Thursday is a day.

Opinion : Thursday was a fun day.

Fact : iPhone is an Apple product.

Opinion : iPhone is good.

- Google searches for facts (currently)
 - Facts can be expressed with topic keywords
 - Google does not search for opinions
 - Opinions are hard to express with keywords
-

Sentiment Analysis



It's a big day & I want to book a table at a nice Japanese restaurant

Seattle has many
★★★★
sushi restaurants

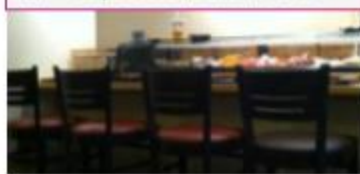


What are people
saying about
the food?
the ambiance?...



Sentiment Analysis

★★★★☆ 428 reviews
\$\$ · Japanese, Sushi Bars



Sample review:

Watching the chefs create incredible edible art made the experience very unique.

My wife tried their ramen and it was pretty forgettable.

All the sushi was delicious!
Easily best sushi in Seattle.

Positive reviews not positive about everything

Experience



Sentiment Analysis



From reviews to topic sentiments

All reviews
for restaurant

★★★★☆ **Reviews**
This is probably my favorite place to eat Japanese in Seattle. My boyfriend and I ordered right of sushi, Japanese omagari (sashimi), and the appetizer rolls and 2 special rolls. I would skip the special rolls, because the right and capture rolls is where this place excels. The rolls, as recommended by other Hoppers was amazing. It's more of a cafe and the atmosphere is the perfect amount of flow for the delicate rolls.

★★★★★ **Reviews**
Dining here at the sushi bar made me feel like sitting front row to an amazing performance. We didn't have much, barged down to the 42 after work, got here breathlessly at 8:15pm, and got the last two seats in the place.

★★★★☆ **Reviews**
I come here having high expectations due to the reviews of this place, but I was bit disappointed. The restaurant is small so do make reservations when you come here. Dishes cost from \$4-10 each and dishes are small.

Novel intelligent
restaurant review app

Experience

★★★★★

Ramen

★★★

Sushi

★★★★★

Easily best sushi
in Seattle.

Examples

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All ▾

Q

Buy Again Browsing History ▾ Cody's Amazon.com Early Black Friday Deals Gift Cards Registry Sell Help

LG Electronics OLED55E8PUA 55-Inch 4K Ultra HD Smart OLED TV (2018... ▸ [Customer reviews](#)

Customer reviews

★★★★☆ 18

3.9 out of 5 stars ▾

5 star		67%
4 star		11%
3 star		5%
2 star		0%
1 star		17%

Write a review



LG Electronics OLED55E8PUA 55-Inch 4K Ultra HD Smart OLED TV (2018 Model)

by LG

Size: 55-inch [Change](#)

Price: \$2,296.99 [prime](#)

Top positive review

[See all 14 positive reviews ▸](#)

 Mayra S. [TOP 1000 REVIEWER](#)

★★★★★ With Google Assistant and new Alpha 9 Processor, 2018 LG Oled's are great upgrades for first time 4K/HDR/Oled Owners

May 3, 2018

Top critical review

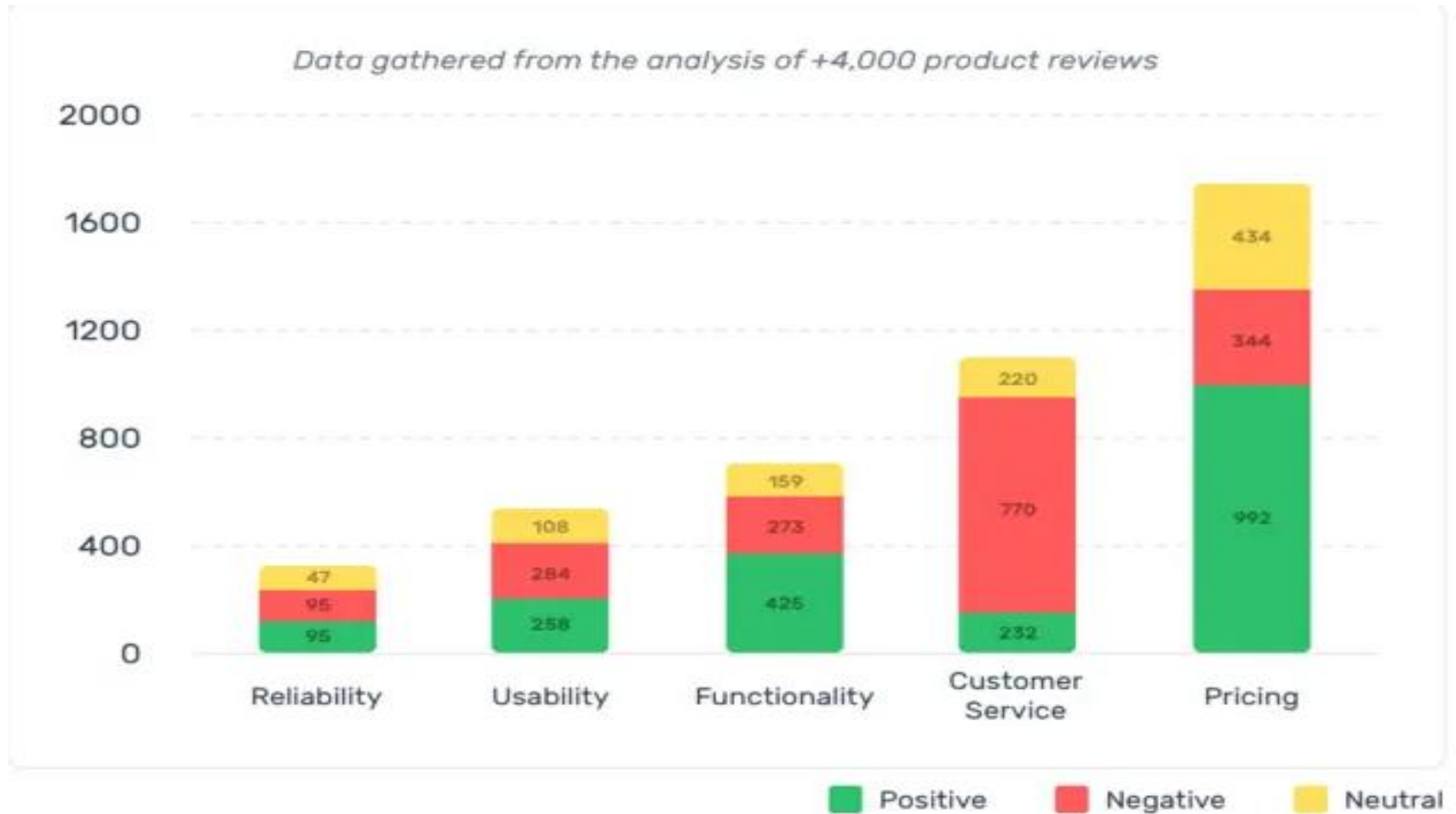
[See all 4 critical reviews ▸](#)

 Brett W.

★★★★☆ Extreme stuttering (no soft transition between frames) is an important factor to consider with OLED TV's

August 3, 2018

Examples



Examples



How people are reacting on bitcoin by analyzing 100 Tweets.



Examples



Sentiment Analytics for the Telecom Company

Negative review

Dear #XYZ there is no network in my area and internet service is pathetic from the past one week. Kindly help me out

-Dated: 10/09/17

Mixed review

Although the value added services being provided are great but the prices are high
#VAS #XYZ

-Dated: 10/09/17

Positive review

Great work done #XYZ Problem resolved by customer care in just one day
#ThanksXYZ

-Dated: 5/06/17



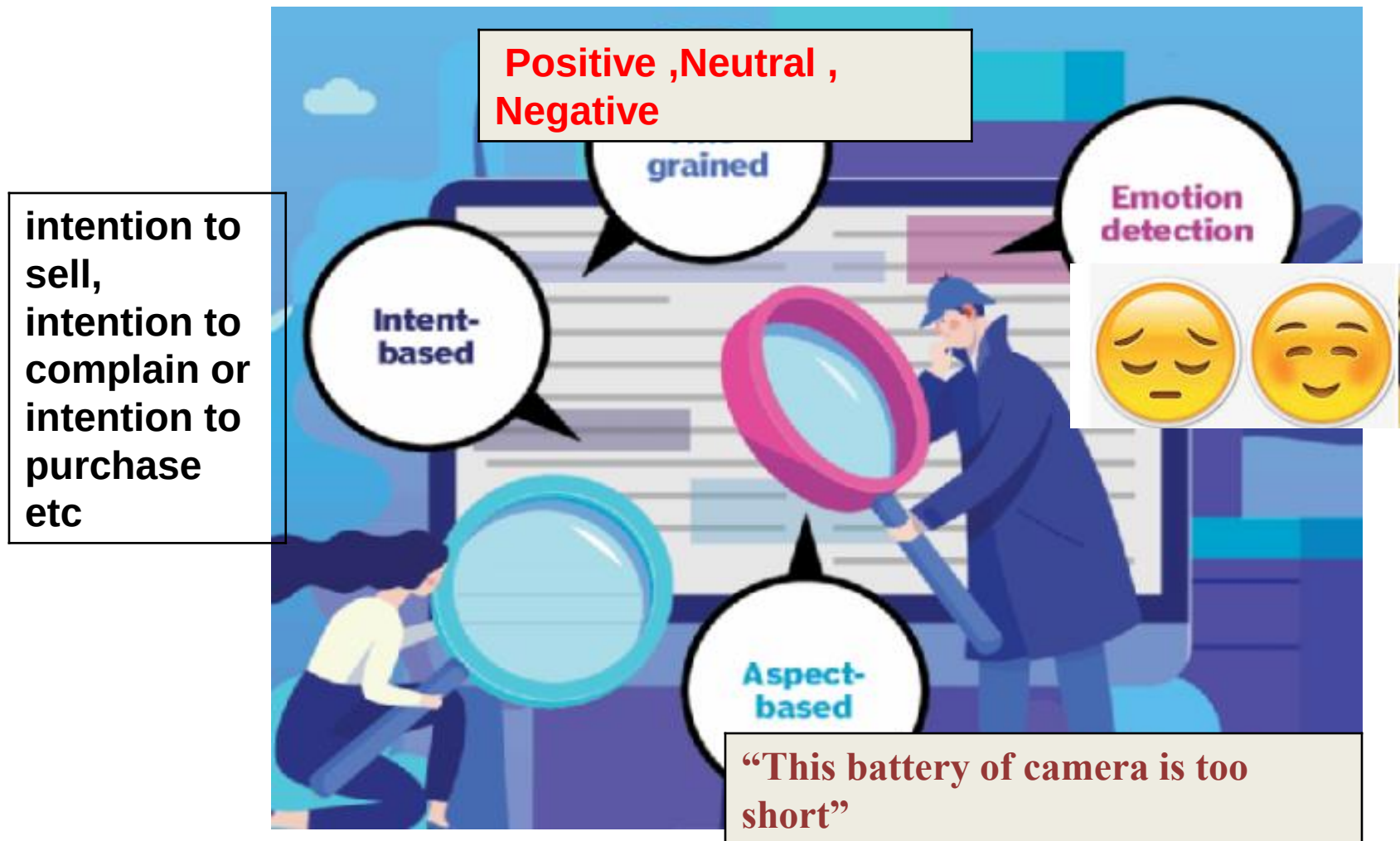
Sentiment Analytics Model

Analyze output to implement **Actionables** like:

- Improve service quality and increase quality checks for locations with max negative reviews.
- Train employees and improve infrastructure of areas with poor service.
- Ensure dedicated customer complaint teams for some areas.
- Tweak the marketing strategy to convey the right message to the customers.

* XYZ depicts name of the telecom company

Types of Sentiment Analysis





What is Sentiment Analysis?

Sentiment Analysis (Opinion Mining) is the computational process of identifying and extracting subjective information from text — determining whether the expressed opinion is Positive, Negative, or Neutral.

Document Level

Classifies overall sentiment of an entire document or review

Sentence Level

Classifies sentiment of individual sentences within text

Aspect Level

Identifies sentiment towards specific entities or attributes

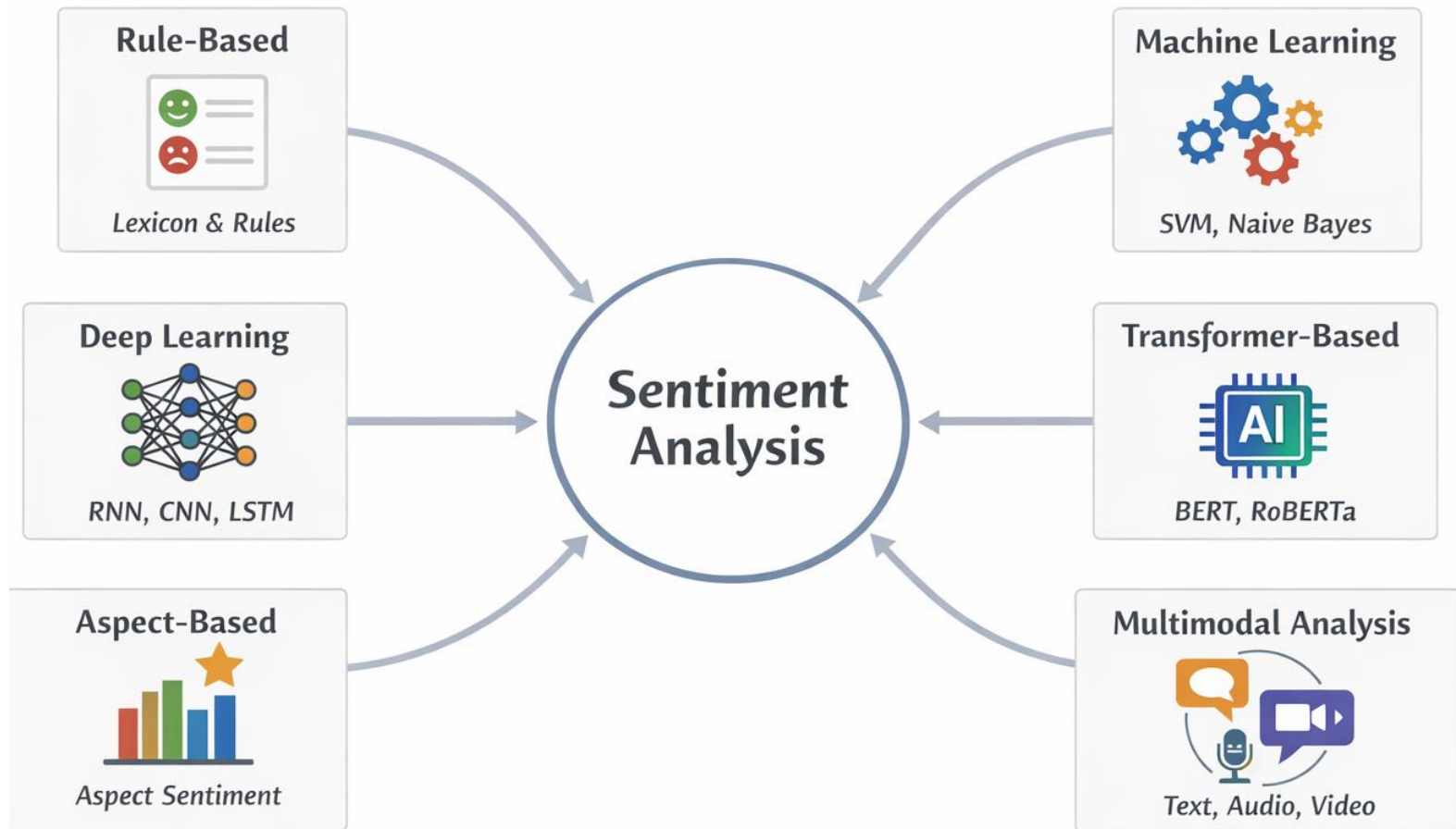
Example:

"The camera quality is amazing but the battery life is terrible." → Aspect-level: [camera: **Positive**] [battery: **Negative**]

Different terms for sentiment analysis



Sentiment Analysis Approaches





Sentiment use cases and approaches

Industry	Sentiment Use Case	Preferred Approach
E-commerce	Review analysis, product ranking	Rule-based / ML/ Transformer / ABSA
Finance	Market mood detection from news & tweets	Transformer models
Healthcare	Patient feedback mining	Rule-based + ML / Transformer
Politics	Public opinion analysis	Deep Learning / Transformer
Customer Support	Ticket prioritization by sentiment	Rule-based + ML / LLM-based
Media & Entertainment	Social media buzz tracking	Rule-based + Transformer hybrid



Rule (Lexicon) Based Methods

How It Works

- 1 Tokenize text into words/phrases
- 2 Look up each token in sentiment lexicon
- 3 Apply valence shifters (negation, intensifiers)
- 4 Aggregate scores → classify polarity

Popular Lexicons

VADER: Social media optimized, handles slang & emojis

SentiWordNet: WordNet-based, covers 117,000 synsets

AFINN: List of 2,477 words scored -5 to +5

Opinion Lexicon: Hu & Liu (2004), ~6,800 words

VADER Example

Input: "The movie was ABSOLUTELY FANTASTIC! Not bad at all 😊"

Output: {neg: 0.0, neu: 0.22, pos: 0.78, compound: **0.87**} → Classification: **POSITIVE** ✓

Rule (Lexicon) Based- Example

- 1.Tokenize
- 2.Remove stop words and punctuations
3. Running the lexicon on the preprocessed data, returns a **positive sentiment** score/measurement because of the presence of a positive word “great” in the input data.

Tokenize	Preprocessing
Sam	
is	stop word
a	stop word
great	
guy	
,	punctuation



Sam	neutral
great	positive
guy	neutral



Sentiment: positive

Machine learning Approach



The song was good .

Tokenization

- The , song, was, good

Classification(Positive, negative ,Neutral)

Apply supervised algorithm

- *Naïve Byes*
 - *Support vector machines*
 - *Maximum Entropy*
-

Lexicons



- Many sentiment applications rely on lexicons to supply features to a model.
- A lexicon is a **resource with information about words**.
- A sentiment lexicon has information such as list of words which are positive and negative.

Example



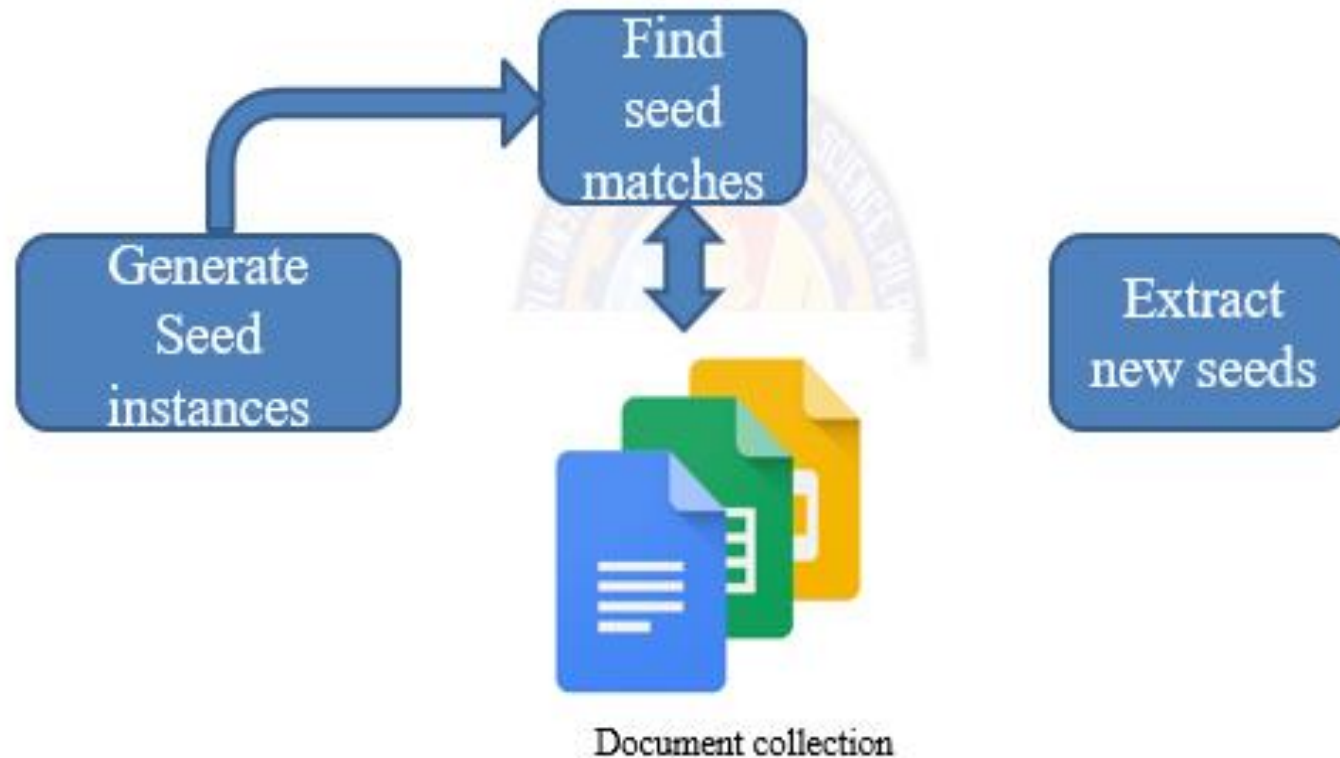
POS	ID	PosScore	NegScore	SynsetTerms	Gloss
a	00001740	0.125	0	able#1	(usually followed by `to') having the necessary means or [...]
a	00002098	0	0.75	unable#1	(usually followed by `to') not having the necessary means or [...]
a	00002312	0	0	dorsal#2 abaxial#1	facing away from the axis of an organ or organism; [...]
a	00002527	0	0	ventral#2 adaxial#1	nearest to or facing toward the axis of an organ or organism; [...]
a	00002730	0	0	acroscopic#1	facing or on the side toward the apex
a	00002843	0	0	basiscopic#1	facing or on the side toward the base
a	00002956	0	0	abducting#1 abducent#1	especially of muscles; [...]
a	00003131	0	0	adductive#1 adducting#1 adducent#1	especially of muscles; [...]
a	00003356	0	0	nascent#1	being born or beginning; [...]
a	00003553	0	0	emerging#2 emergent#2	coming into existence; [...]



Corpus based lexicon generator

- A more sophisticated technique is a corpus-based approach which relies on syntactic or **co-occurrence patterns** together with a seed list of opinion words.
- The technique **starts with a list of seed opinion adjective words**, and uses them and a **set of linguistic constraints** or conventions on connectives to **identify additional adjective opinion words and their orientations**.

Bootstrapping architecture



Example

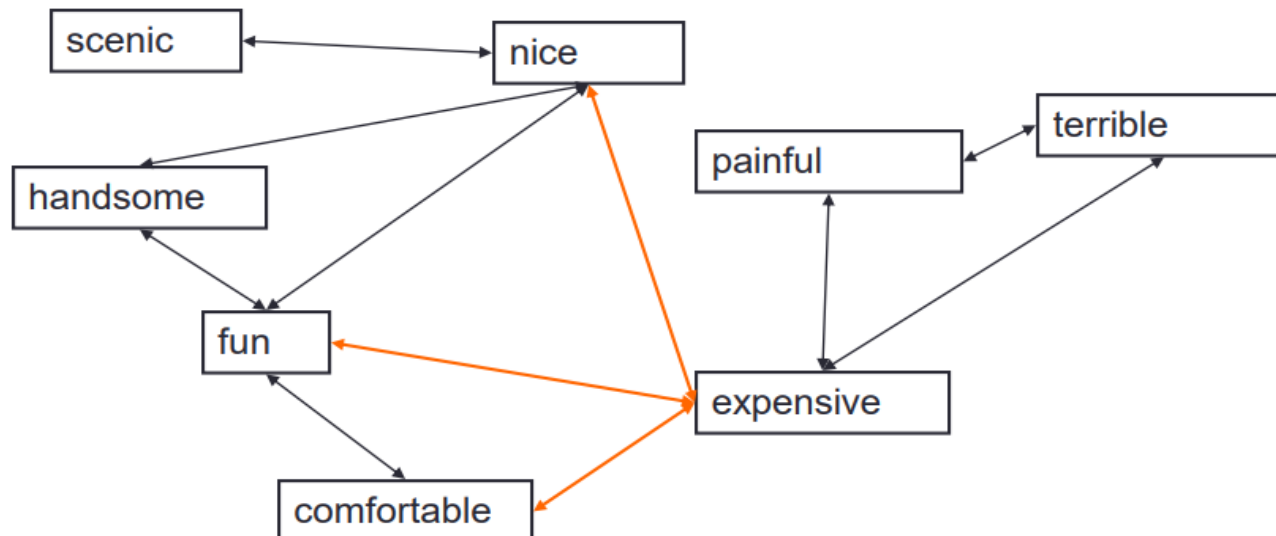


- Adjectives conjoined by “and” have same polarity
Fair and legitimate ,corrupt and brutal
- Adjectives conjoined by “but” do not
Fair but brutal

Algorithm



1. Generate a Labeled seed set of adjectives
2. Expand seed set to conjoined adjectives by looking up in a corpus/web search
3. builds a graph of adjectives linked by the same or different semantic orientation



Turney algorithm



- Extract a phrasal lexicon from reviews
- Learn polarity of each phrase
- Rate a review by the average polarity of its phrases

First Word	Second Word	Third Word (not extracted)
JJ	NN or NNS	anything
RB, RBR, RBS	JJ	Not NN nor NNS
JJ	JJ	Not NN or NNS
NN or NNS	JJ	Nor NN nor NNS
RB, RBR, or RBS	VB, VBD, VBN, VBG	anything

Two-word phrases with adjectives



How to measure polarity of a phrase

- Positive phrases co-occur more with “excellent”
 - Negative phrases co-occur more with “poor”
 - But how to measure co-occurrence?
-



Pointwise Mutual Information

- Pointwise mutual information: How much more do events x and y co-occur than if they were independent?

$$\text{PMI}(\text{word}_1, \text{word}_2) = \log_2 \frac{P(\text{word}_1, \text{word}_2)}{P(\text{word}_1)P(\text{word}_2)}$$

- If two words are **statistically independent**, $\text{PMI}=0$
- If two words tend to not at all **co-occur**, **PMI is negative**
- If two words tend to co-occur, **PMI is positive**
- Does phrase appear more with “poor” or “excellent”?
 - $\text{Polarity}(\text{phrase}) = \text{PMI}(\text{phrase}, \text{"excellent"}) - \text{PMI}(\text{phrase}, \text{"poor"})$

Two reviews for Positive and Negative phrases



Phrase	POS tags	Polarity
online service	JJ NN	2 . 8
online experience	JJ NN	2 . 3
direct deposit	JJ NN	1 . 3
local branch	JJ NN	0 . 42
...		
low fees	JJ NNS	0 . 33
true service	JJ NN	-0 . 73
other bank	JJ NN	-0 . 85
inconveniently located	JJ NN	-1 . 5
<i>Average</i>		0 . 32

Phrase	POS tags	Polarity
direct deposits	JJ NNS	5 . 8
online web	JJ NN	1 . 9
very handy	RB JJ	1 . 4
...		
virtual monopoly	JJ NN	-2 . 0
lesser evil	RBR JJ	-2 . 3
other problems	JJ NNS	-2 . 8
low funds	JJ NNS	-6 . 8
unethical practices	JJ NNS	-8 . 5
<i>Average</i>		-1 . 2



Wordnet based polarity estimation

- WordNet: online thesaurus indexing words by synonyms
 - Create positive (“good”) and negative seed-words (“terrible”)
 - Find Synonyms and Antonyms
 - Positive Set: Add synonyms of positive words (“well”) and antonyms of negative words
 - Negative Set: Add synonyms of negative words (“awful”) and antonyms of positive words (“evil”)
 - Repeat, following chains of synonyms
 - Filter
-

Aspect Based Sentiment Analysis (ABSA)



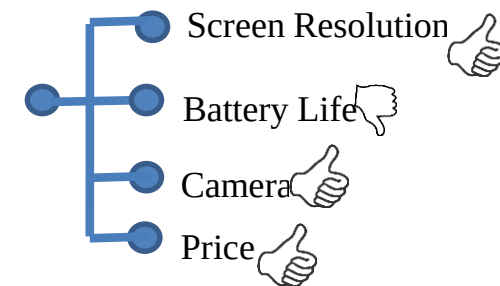
“(1) I bought an *iPhone* a few days ago. (2) It was such a *nice phone*. (3) The *touch screen* was really *cool*. (4) The *voice quality* was *clear* too. (5) Although the *battery life* was *not long*, that is ok for me. (6) However, *my mother* was mad with me as I did not tell her before I bought it. (7) She also thought the *phone* was too *expensive*, and wanted me to return it to the shop. ...”

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Aspect Based Sentiment Analysis (ABSA)

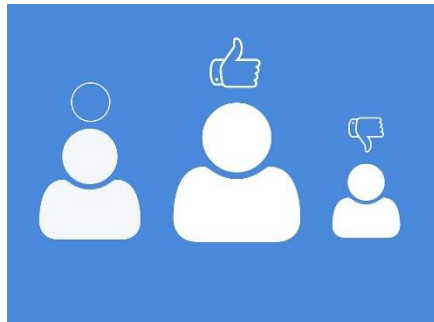


- Each opinion is defined as **quintuple** (e, a, s, h, t) , where e is an entity and a is one of its aspects, s is the sentiment on the aspect a , h is the opinion holder and t is the time when the opinion is expressed.
- Find the **target(Aspect/Entity)** of the sentiment.
- Two approaches
 - Find most common noun phrases
 - Build a classifier



Frequency-Based Aspect Extraction

- A key characteristic is that an **opinion always has a target**.
- Exploit **syntactic structures** to depict opinion and target relationships



Review corpus



Association Rule
Mining

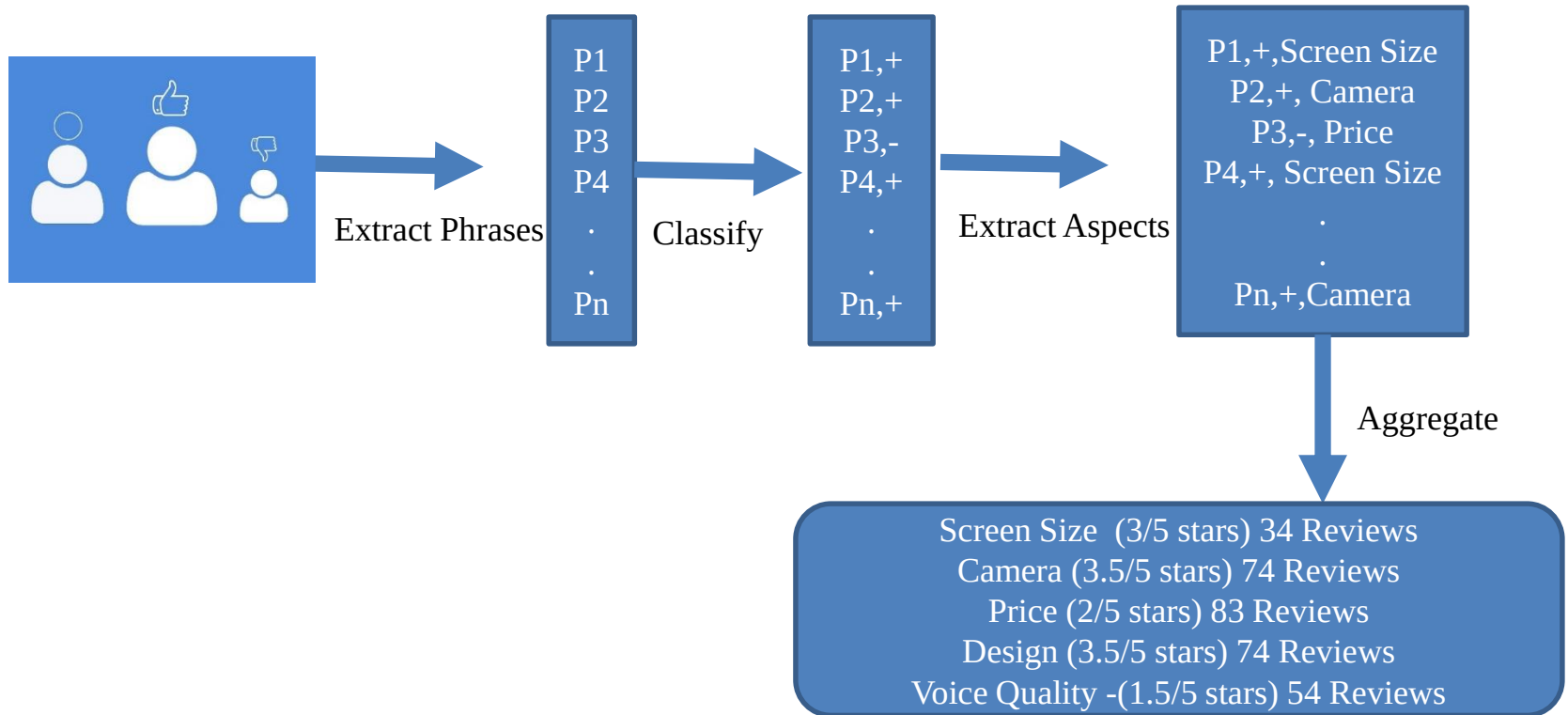
Screen Size – 100/500
Camera Resolution – 300/500
Battery Life – 350/500
Price -450/500
Voice clarity – 325/500

Examples of aspects extracted

- Those **candidate aspects with the highest frequency counts** are almost always the most **important aspects of the product**.
- Assumption: Corpus has **reasonable number of reviews** and **belong to same product**.

Entity	Aspects extracted
Casino	Casino, buffet, pool, resort, beds
Department store	Selection, department, sales, shop, clothing
Greek Restaurant	Food, Wine, Service, Appetizer, lamb

Architecture for ABSA



Blair-Glodensohn from Google

Machine Learning Methods



Key Algorithms

- Naive Bayes** Fast, interpretable baseline; strong for text
- Logistic Regression** Probabilistic, works well with TF-IDF
- SVM** High-dimensional, excellent margin classifier
- Random Forest** Ensemble, handles feature interactions
- Gradient Boosting** XGBoost/LightGBM — state-of-art classic ML

Feature Engineering

- Bag of Words (BoW):** Word frequency vectors, sparse
- TF-IDF:** Term frequency $\times$ inverse document freq.
- N-grams:** Captures bi/trigram context ('not good')
- Word Embeddings:** Word2Vec, GloVe — dense semantic vectors
- POS Tags:** Part-of-speech as additional features

Deep Learning & Transformers

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Architecture Evolution



BERT for Sentiment (Fine-tuning)

- ◆ Pre-trained on 3.3B words (BooksCorpus + Wikipedia)
- ◆ Add [CLS] token → fine-tune on labelled sentiment data
- ◆ Bidirectional context captures long-range dependencies
- ◆ State-of-art: ~96% accuracy on SST-2 benchmark

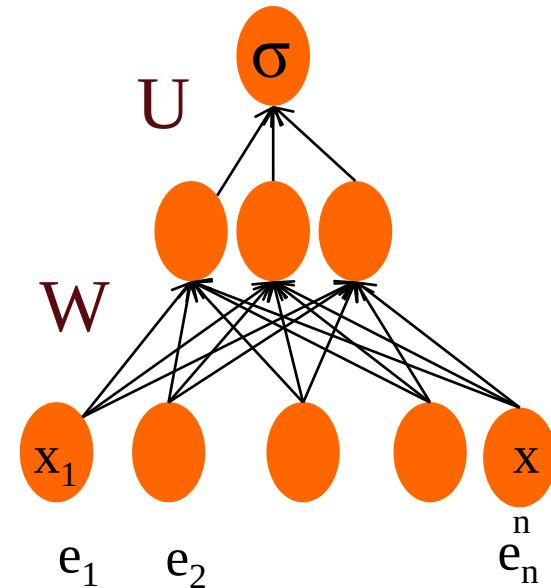
Accuracy Comparison

VADER (lexicon)	79%
Naive Bayes	83%
SVM + TF-IDF	88%
BERT fine-tuned	96%

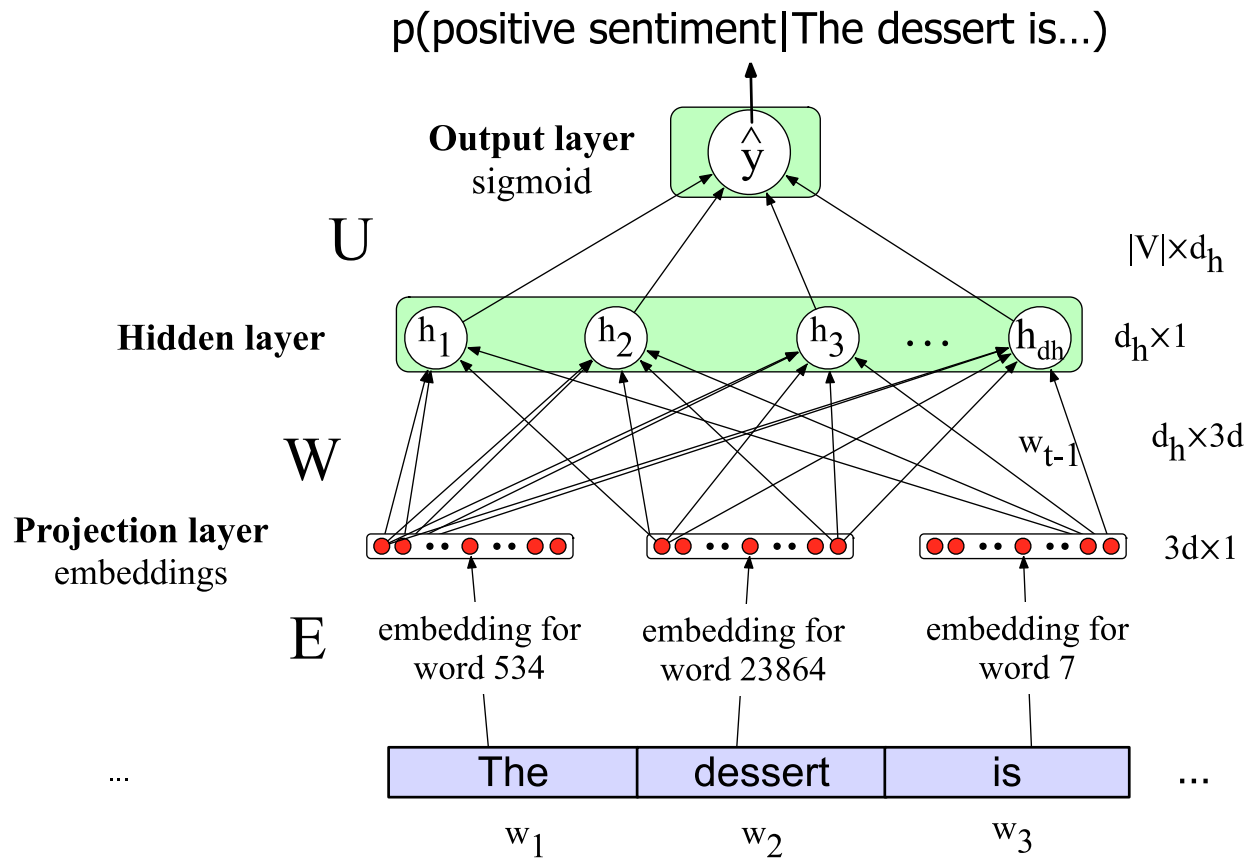
💡 **LLM Zero-shot:** GPT-4 prompt: "Classify sentiment as Positive/Negative/Neutral and return JSON with confidence score" — achieves 91% without any fine-tuning, ideal when labelled data is scarce.

Feedforward nets for Sentiment Analysis

- The real power of deep learning comes from the ability to **learn** features from the data
- Instead of using hand-built human-engineered features for classification
- Use learned representations like embeddings!



Neural Net Classification with embeddings as input features!





Twitter Product Sentiment at Scale

Scenario: Consumer Electronics Brand — iPhone 15 Launch

2.4M

Tweets Analysed

72hrs

Data Collection Window

89%

Model Accuracy

94ms

Avg Inference Latency

Key Findings

- 60% Positive sentiment around camera improvements
- 28% Negative around USB-C port transition issues
- Sentiment spike: -0.42 after competitor price drop announcement
- Influencer posts with >10k followers: 3.2× sentiment amplification

Architecture Used

Ingestion: Twitter API v2 → Kafka streams

Preprocessing: SpaCy — tokenize, clean, normalize

Model: RoBERTa-base fine-tuned on TweetEval

Serving: TorchServe on AWS ECS (autoscale)

Dashboard: Kibana + Elasticsearch real-time viz

Tools & Technology Stack



Data Collection	Preprocessing	Modelling	Serving / MLOps	Monitoring & Viz
● Tweepy (Twitter API) ● BeautifulSoup / Scrapy ● NewsAPI, Reddit PRAW ● AWS Kinesis / Kafka	● NLTK, SpaCy ● Hugging Face Tokenizers ● TextBlob ● ftfy (unicode fix)	● Hugging Face Transformers ● PyTorch / TensorFlow ● scikit-learn ● VADER / TextBlob	● FastAPI + TorchServe ● MLflow, DVC ● Kubernetes + Docker ● AWS SageMaker	● Grafana + Prometheus ● Kibana / ElasticSearch ● Weights & Biases ● Evidently AI



Fine-Grained Opinion Mining

Aspect-Based Sentiment Analysis

ABSA identifies sentiments tied to specific aspects within text, offering detailed insight beyond overall sentiment.

Applications of ABSA

ABSA aids product development and customer experience by revealing strengths and weaknesses in feedback.

Implementation Techniques

Modern ABSA uses transformer models and NLP pipelines to extract and classify aspects and sentiments efficiently.

Benefits and Challenges

While complex to design, ABSA provides significantly richer insights for decision-making than standard sentiment analysis.

Industry Applications



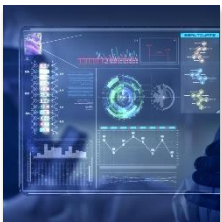
Finance Sector Impact

Sentiment analysis helps financial companies understand customer emotions and improve call center service efficiency.



Airline Industry Usage

Airlines analyze customer feedback and sentiment trends to enhance service quality and operational performance.



Business Value of Sentiment Analysis

Sentiment analysis drives data-driven decisions improving customer experience and competitive advantage across industries.



Application Areas Summary

- Businesses and organizations: interested in opinions
 - product and service benchmarking
 - market intelligence
 - survey on a topic
 - Individuals: interested in other's opinions when
 - Purchasing a product
 - Using a service
 - Tracking political topics
 - Other decision making tasks
 - Ads placements: Placing ads in user-generated content
 - Place an ad when one praises an product
 - Place an ad from a competitor if one criticizes a product
 - Opinion search: providing general search for opinions
 - Text-driven forecasting: insights about other areas from text
-

Production Environment Issues & Challenges



Data Drift & Concept Drift

Language evolves — slang, new entities, cultural shifts. Model trained on 2020 tweets degrades on 2024 data.

Solution: Continuous retraining pipelines, Evidently AI monitoring, sliding-window retraining.

Domain Mismatch

Model trained on movie reviews fails on medical forums ('positive biopsy' $\neq$ positive sentiment).

Solution: Domain-specific fine-tuning; domain-adaptive pretraining (DAPT).

Sarcasm & Irony

"Great, another system outage" — lexicon scores this as +0.8 (positive). Hard even for humans.

Solution: Contextual models (BERT), multi-modal signals, discourse-level features.

Multilingual & Code-Switching

"This product is shabby but yaar dil se acha hai" — mixed Hindi/English (Hinglish).

Solution: mBERT, XLM-RoBERTa, language detection + routing pipelines.

Scalability & Latency

BERT inference: ~50ms/sample; 1M tweets/day = 13.8 hrs on single GPU.

Solution: Model quantization (INT8), distillation (DistilBERT 60% faster), batch inference, async queues.

Class Imbalance

Typical distribution: 65% neutral, 25% positive, 10% negative. Model biased toward majority.

Solution: SMOTE oversampling, class-weighted loss, threshold tuning per class.

Evaluation Measures

—Metrics



Confusion Matrix (Binary)

	Pred +	Pred -
Actual +	TP = 850	FN = 50
Actual -	FP = 30	TN = 520

Accuracy

$$(TP + TN) / (TP + TN + FP + FN)$$
$$= (850 + 520) / 1450 = \mathbf{0.945 = 94.5\%}$$

Precision

$$TP / (TP + FP)$$
$$= 850 / (850 + 30) = \mathbf{0.966 = 96.6\%}$$

Recall (Sensitivity)

$$TP / (TP + FN)$$
$$= 850 / (850 + 50) = \mathbf{0.944 = 94.4\%}$$

F1-Score

$$2 \times (Precision \times Recall) / (Precision + Recall)$$
$$= 2 \times (0.966 \times 0.944) / (0.966 + 0.944) = \mathbf{0.955 = 95.5\%}$$

Evaluation Measures

—Metrics



AUC-ROC Curve

AUC = 0.97

Area Under the ROC Curve measures model discrimination at all classification thresholds. AUC = 1.0 → perfect; AUC = 0.5 → random.

Formula: $AUC = \int_0^1 TPR(FPR^{-1}(t)) dt$

Our model AUC = 0.97 means 97% probability of correctly ranking a positive instance above a negative one.

Matthews Correlation Coefficient (MCC)

$$MCC = (TP \times TN - FP \times FN) / \sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}$$

With TP=850, TN=520, FP=30, FN=50:

$$\begin{aligned} &= (850 \times 520 - 30 \times 50) / \sqrt{(880)(900)(550)(570)} \\ &= (442000 - 1500) / \sqrt{(248,832,000,000)} \\ &= 440500 / 498,831 \approx 0.883 \end{aligned}$$

MCC = 0.883 → Strong correlation (range: -1 to +1)

Cohen's Kappa (Inter-Annotator Agreement)

$$\kappa = (P_o - P_e) / (1 - P_e)$$

Where P_o = observed agreement, P_e = expected agreement by chance

Example: Two annotators agree on 87% of samples, expected chance agreement = 52%

$$\kappa = (0.87 - 0.52) / (1 - 0.52) = 0.35 / 0.48 \approx 0.73 \rightarrow$$

Substantial agreement

Cross-Entropy Loss (Training Objective)

$$L = -(1/N) \sum_i \sum_j y_{ij} \cdot \log(\hat{p}_{ij})$$

For 3-class (Pos/Neg/Neu) with sample [1,0,0] predicted as [0.85, 0.10, 0.05]:

$$\begin{aligned} L &= -(1 \cdot \log(0.85) + 0 \cdot \log(0.10) + 0 \cdot \log(0.05)) \\ &= -\log(0.85) = 0.163 \end{aligned}$$

Lower loss = better calibrated probabilities

Approach Comparison & When to Use



Approach	Accuracy	Speed	Labelled Data	Best For
Lexicon-Based	Low–Med (79%)	⚡ Very Fast	None	Real-time, no training budget
Classical ML	Medium (83–88%)	Fast	Required	Tabular data, interpretability
Deep Learning CNN/LSTM	High (90–93%)	Moderate	Large	Complex sequential data
BERT Fine-tuned	Very High (94–96%)	Slower	Moderate	High-accuracy production use
LLM Zero-shot	High (91%)	Slow/API	None	Low-data, multi-task scenarios

🏆 **Recommendation:** Start with VADER for rapid prototyping. Use a fine-tuned RoBERTa/BERT for production. Combine LLM zero-shot for low-resource domains. Always monitor for drift with Evidently AI or WhyLogs.



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